ESD Protection When Handling Inverter Boards

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Effective date	2025-08-01	**Owner**	Quality Engineering
Classification	Internal		

1 Purpose

Inverter control boards are sensitive to electrostatic discharge (ESD). Damage is often invisible and only shows up later as a field failure. This instruction applies to everybody who handles boards in the electronic pre-assembly area (EPA).

2 Rules in the EPA

- Only enter the EPA with ESD shoes or heel straps.
- Wear the wrist strap whenever you touch a board.
- Keep boards in ESD bags or ESD trays. Normal plastic, paper folders and personal items are not allowed on the workbench.
- The relative humidity in the EPA must be between 30 % and 70 %. The display is next to the door. If it is outside this range, inform the shift leader.

3 Wrist strap check

Test your wrist strap at the start of each shift on the tester at the EPA entrance. The tester passes the strap if the total resistance is between 0.75 megohm and 35 megohm. The strap contains a 1 megohm safety resistor. If the tester shows FAIL, take a new strap from the cabinet and test again. Never work with a strap that failed the test.

4 Handling boards

Hold boards only at the edges. Do not touch components or connectors. Boards that fell on the floor are scrapped, even if they look fine.

5 ESD incidents

If you notice that you worked without a tested strap, or a board was handled outside the EPA, inform the shift leader. The affected boards are blocked and checked by quality.

6 Revision history

Rev	Date	Change
A	2024-05-20	First issue
B	2025-08-01	Humidity range added, dropped boards are scrapped